LEVEL UP ACADEMY

Class 10 | CBSE | Science

ANSWER KEY & MARKING SCHEME

Drill: Assertion–Reason & Case-Based Questions (CBSE)

Chapter / Topic: Mixed chapters · Week 9 of the 90-day plan

For teacher use only. Do not distribute to students.

PART A: ASSERTION–REASON

Q1. **A:** The colour of copper sulphate solution changes when a zinc rod is dipped in it. **R:** Zinc is more reactive than copper.

Answer: (a)

Q2. **A:** Calcium carbonate decomposes on heating. **R:** Decomposition reactions are generally exothermic.

Answer: (c). Most decomposition reactions are endothermic: they need heat, light or electricity.

Q3. **A:** Pure water is neutral. **R:** Pure water has equal concentrations of H^+ and OH^- ions.

Answer: (a)

Q4. **A:** Sodium is stored under kerosene. **R:** Sodium is a non-metal.

Answer: (c). Sodium is a highly reactive metal. It is stored under kerosene because it reacts vigorously with air and moisture.

Q5. **A:** Ionic compounds have high melting points. **R:** Ionic compounds are soluble in water.

Answer: (b). Both are true. High melting points are due to the strong forces between ions.

Q6. **A:** Diamond is a good conductor of electricity. **R:** In diamond, each carbon atom is bonded to four other carbon atoms.

Answer: (d). Diamond is a poor conductor; graphite conducts.

Q7. **A:** The walls of capillaries are only one cell thick. **R:** Materials are exchanged between the blood and the surrounding cells through the capillaries.

Answer: (a)

Q8. **A:** A deficiency of insulin causes diabetes. **R:** Insulin is secreted by the thyroid gland.

Answer: (c). Insulin is secreted by the pancreas.

Q9. **A:** In human beings, the sex of the child is decided by the father. **R:** Half the sperm carry an X chromosome and half carry a Y chromosome.

Answer: (a)

Q10. **A:** A concave lens always forms a virtual image of a real object. **R:** The power of a concave lens is negative.

Answer: (b). Both are true, but the image is virtual because a concave lens diverges the rays.

Q11. **A:** The resistance of a wire decreases when its length is increased. **R:** The resistance of a wire is directly proportional to its length.

Answer: (d). Resistance increases with length.

Q12. **A:** Plants are called producers. **R:** Plants release oxygen during photosynthesis.

Answer: (b). Both are true, but plants are producers because they make food from inorganic substances using sunlight.

Q13. **A:** Copper reacts with dilute hydrochloric acid to give hydrogen gas. **R:** Copper is less reactive than hydrogen.

Answer: (d). Copper does not displace hydrogen from dilute acids.

Q14. **A:** Bile juice does not contain any enzyme. **R:** Bile is produced by the liver.

Answer: (b). Both are true, but R does not explain why bile has no enzymes. Bile's role is to make the medium alkaline and emulsify fats.

PART B: CASE-BASED QUESTIONS

Q15. Ethanol is used in medicines, as a solvent and as a fuel. Even small amounts of methanol are poisonous: in the liver it is oxidised to methanal,… **[4]**

(i) Ethanol made unfit for drinking by adding poisonous substances such as methanol, and coloured with a dye. [1]

(ii) In the liver it is oxidised to methanal, which can damage the optic nerve and cause blindness, or even death. [1]

(iii) $2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2$. Bring a burning matchstick near the gas. It burns with a pop sound, which shows it is hydrogen. [2]

Q16. In tissue culture, new plants are grown by taking a few cells or a small piece of tissue from the growing tip of a plant. The tissue is placed in an… **[4]**

(i) A small, unorganised mass of rapidly dividing cells formed from the tissue in the culture medium. [1]

(ii) Asexual (a form of vegetative propagation). [1]

(iii) (Any two) Many plants can be produced from very little parent tissue. All the plants are genetically like the parent, so good characters are kept. It is fast, and does not depend on seeds or seasons. It can produce disease-free plants. [2]

Q17. The resistivity (in Ω m, at 20 °C) of some materials is: silver 1.60×10^{-8}, copper 1.62×10^{-8}, aluminium 2.63×10^{-8}, tungsten 5.20×10^{-8} and… **[4]**

(i) Silver, which has the lowest resistivity. [1]

(ii) Glass [1]

(iii) Nichrome has a very high resistivity, so it produces a lot of heat, and it does not oxidise (burn) easily at high temperatures. Copper has a very low resistivity, so very little energy is wasted as heat in the wires. [2]

Q18. A student traces the path of a ray of light through a rectangular glass slab. The ray bends towards the normal when it enters the glass, and away… **[4]**

(i) Light slows down in glass, which is optically denser than air. [1]

(ii) The two refracting faces are parallel, so the bending at the second face exactly undoes the bending at the first face. [1]

(iii) The lateral displacement would increase. A refractive index of 1.5 means light travels 1.5 times faster in vacuum than in glass ($v = c/1.5 = 2 \times 10^8$ m/s). [2]